HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2023)

Student No.: 11 Name: _____ Marks: / 38

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. Let $h(x) = ax^6 + 16x^3 + b$, where a and b are constants. If $h(x)$ is divisible by $2x - 3$, find the remainder when $h(x)$ is divided by $2x + 3$.

   A. -108

   B. -54

   C. 54

   D. 108

(DSE-2023 PII Q09)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

2. It is given that z varies as the square of x and the cube root of y. When $x = 12$ and $y = 64$, $z = 36$. When $x = 16$ and $y = 729$, $z =$

A. 108.

B. 144.

C. 162.

D. 216.

(DSE-2023 PII Q13)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

3. The length of a side of a solid cube is $60\,\text{cm}$. The volume of a solid right circular cylinder is equal to the volume of the cube while the curved surface area of the circular cylinder is equal to the total surface area of the cube. Find the base radius of the circular cylinder.

A. $20\,\text{cm}$

B. $30\,\text{cm}$

C. $76\,\text{cm}$

D. $172\,\text{cm}$

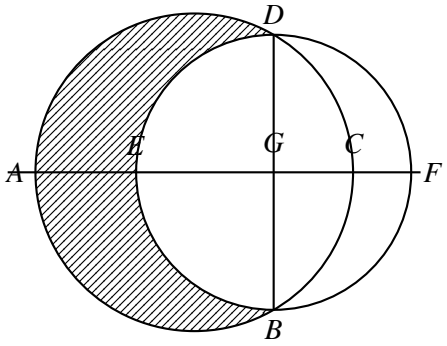
(DSE-2023 PII Q15)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

4. In the figure, AC is a diameter of the circle $ABCD$ while BD and EF are diameters of the circle $BEDF$. It is given that C and E lie on AF. Let G be the point of intersection of AF and BD. If $AG = 30\,\text{cm}$ and $CG = 10\,\text{cm}$, find the area of the shaded region correct to the nearest cm^2.



A. $209\,\text{cm}^2$

B. $367\,\text{cm}^2$

C. $383\,\text{cm}^2$

D. $540\,\text{cm}^2$

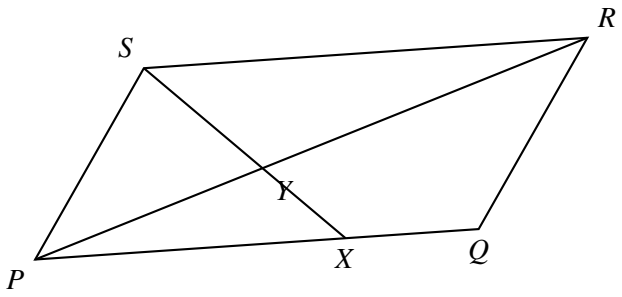
(DSE-2023 PII Q16)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

5. In the figure, $PQRS$ is a parallelogram. Let X be a point lying on PQ. Denote the point of intersection of PR and SX by Y. If the area of $\triangle PXY$ and the area of the quadrilateral $QRYX$ are $32\,\text{cm}^2$ and $58\,\text{cm}^2$ respectively, then the area of $\triangle RSY$ is



A. $40\,\text{cm}^2$.

B. $50\,\text{cm}^2$.

C. $58\,\text{cm}^2$.

D. $72\,\text{cm}^2$.

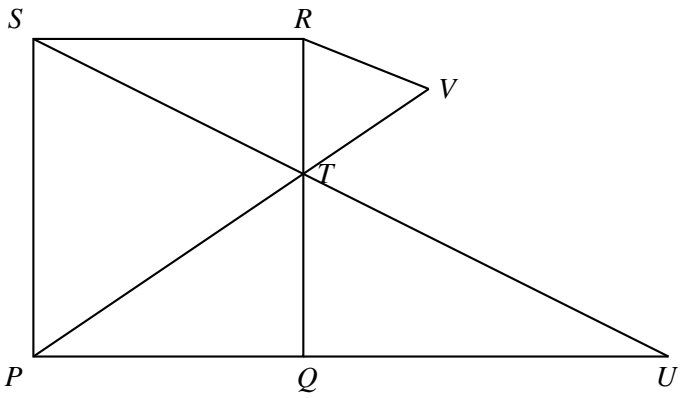
(DSE-2023 PII Q17)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

6. In the figure, $PQRS$ is a rectangle. Let T be a point lying on QR such that $\angle PTS = 90°$. PQ produced and ST produced meet at the point U. PT is produced to the point V such that $RT = RV$. Which of the following must be true?



A. $RV \parallel ST$

B. $\angle PTQ = \angle RTS$

C. $\triangle PST \sim \triangle UTQ$

D. $\triangle PQT \cong \triangle TRS$

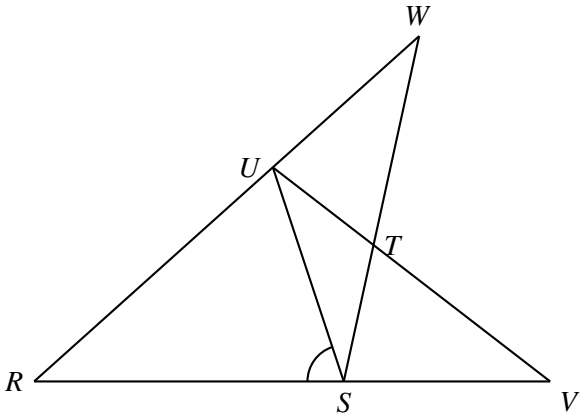
(DSE-2023 PII Q21)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

7. The figure shows the cyclic quadrilateral $RSTU$, where $ST = TU$. RS produced and UT produced meet at the point V while RU produced and ST produced meet at the point W. If $\angle RWS = 32^\circ$ and $\angle RVU = 48^\circ$, then $\angle RSU =$



A. 65°.

B. 73°.

C. 80°.

D. 82°.

(DSE-2023 PII Q22)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

8. The rectangular coordinates of the point P are $(\sqrt{2}, -\sqrt{2})$. If P is rotated anticlockwise about the origin through $90°$, then the polar coordinates of its image are

A. $(\sqrt{2}, 45°)$.

B. $(\sqrt{2}, 225°)$.

C. $(2, 45°)$.

D. $(2, 225°)$.

(DSE-2023 PII Q24)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

9. The equations of the straight lines ℓ and L are $9x+12y-37=0$ and $12x+16y+85=0$ respectively. ℓ cuts the x-axis at the point A while L cuts the y-axis at the point B. Let P be a moving point in the rectangular coordinate plane such that the perpendicular distance from P to ℓ is equal to the perpendicular distance from P to L. Denote the locus of P by Γ. Which of the following are true?

I. Γ is parallel to L.

II. Γ is perpendicular to AB.

III. Γ passes through the mid-point of AB.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2023 PII Q26)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

10. It is given that $\log_8 y$ is a linear function of $\log_4 x$. The intercepts on the vertical axis and on the horizontal axis of the graph of the linear function are 5 and 3 respectively. Which of the following must be true?

A. $x^5y^2 = 8^{10}$

B. $x^6y^5 = 8^{20}$

C. $x^{10}y^3 = 8^{20}$

D. $x^9y^{10} = 8^{30}$

(DSE-2023 PII Q33)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

11. Consider the following system of inequalities:

$$\begin{cases} x+3 \geq 0 \\ 2x+3y-12 \leq 0 \\ 5x-3y+12 \leq 0 \end{cases}$$

Let D be the region which represents the solution of the above system of inequalities. Find the range of values of β such that the greatest value of $\beta x+6y$ is 24, where (x, y) is a point lying in D.

A. $\beta \leq -10$

B. $\beta \geq -10$

C. $\beta \leq 4$

D. $\beta \geq 4$

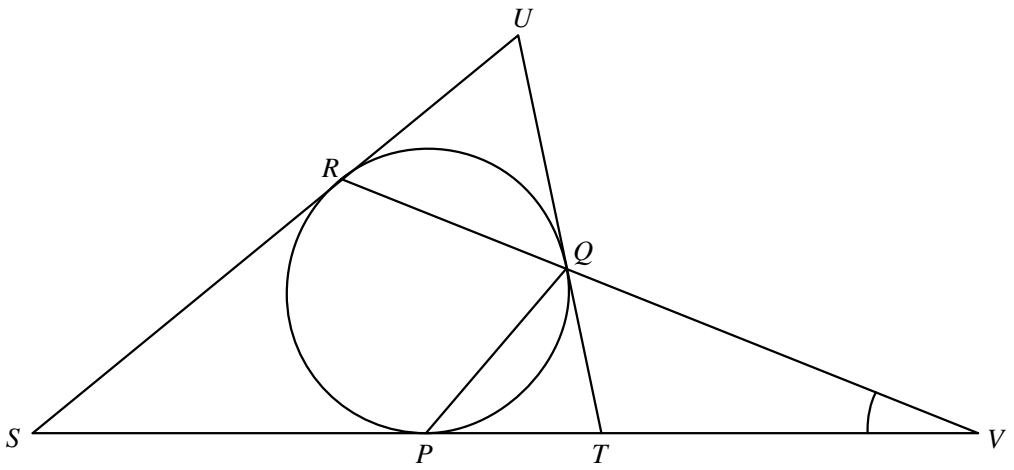
(DSE-2023 PII Q37)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

12. In the figure, P, Q and R are points lying on a circle. ST, TU and SU are the tangents to the circle at P, Q and R respectively. RQ produced and ST produced meet at the point V. If $\angle PSR = 34^\circ$ and $\angle QPT = 46^\circ$, then $\angle PVQ =$



A. 17°.

B. 22°.

C. 27°.

D. 28°.

(DSE-2023 PII Q38)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

13. The straight line $hx + ky = 6$ and the circle $x^2 + y^2 - 8x - 4y - 18 = 0$ intersect at the points M and N, where h and k are constants. If the coordinates of the mid-point of MN are $(1, 0)$, find k.

A. 4

B. 6

C. 9

D. 12

(DSE-2023 PII Q39)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

14. The base of the right pyramid $VABCD$ is the square $ABCD$. Let θ be the angle between $\triangle ABV$ and $\triangle BCV$. If $AB : AV = 5 : 4$, then $\cos\theta =$

A. $\dfrac{-25}{39}$.

B. $\dfrac{-17}{33}$.

C. $\dfrac{-9}{16}$.

D. 0.

(DSE-2023 PII Q40)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

15. The equations of the straight lines L_1 and L_2 are $3x-4y+k=0$ and $4x+3y-k=0$ respectively, where k is a positive constant. It is given that L_1 cuts the x-axis at the point P. Denote the point of intersection of L_1 and L_2 by Q. If R is a point lying on L_2 such that the in-centre of $\triangle PQR$ lies on the x-axis, then the x-coordinate of R is

A. $-7k$.

B. $-k$.

C. k.

D. $7k$.

(DSE-2023 PII Q41)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

16. There are 15 teachers in a group. If 5 teachers are selected from the group to form a committee consisting of 1 chairperson and 4 members, how many different committees can be formed?

A. 3003

B. 15015

C. 20475

D. 360360

(DSE-2023 PII Q42)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

17. When a boy throws a dart, the probability that he hits the target is 0.6 . If this boy throws the dart 4 times, find the probability that he hits the target at least 2 times.

A. 0.5248

B. 0.7056

C. 0.8208

D. 0.8464

(DSE-2023 PII Q43)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

18. The table below shows the scores (in marks) and the corresponding standard scores of three students in an examination.

Score (marks)	46	x	86
Standard score	-3	1	2

Find x.

A. 64

B. 66

C. 70

D. 78

(DSE-2023 PII Q44)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

19. It is given that n is an integer. Let u, v and w be the standard deviation, the median and the range of the group of numbers $\{1-9n, 3-9n, 4-9n, 5-9n, 7-9n\}$ respectively. Which of the following must be true?

I. $u = 2$

II. $v < 4$

III. $w > 6$

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2023 PII Q45)

I can finish _____ questions

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.